

V3 empirical study · Instruction

Six models · three underlyings · four volatility regimes · 1.5-year monthly calibration

What this report is

- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of six American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and volatility regimes?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Mark column says BEST.

Experimental design (held fixed for every model)

- Models: GBM, GARCH(1,1), Heston (no jumps), Merton jump-diffusion, GARCH-Merton, Heston-Merton (Bates).
- Companies: SPY (primary), AAPL, MSFT. Each name is calibrated and priced separately.
- Regimes (V3 one-year evaluation windows): Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- The late-cycle and COVID files share September 2019 (V3 notebook dates). All other regime days are disjoint. Contracts are built independently in each window.
- Calibration window: 1.5-year ending at each monthly update. Rolling: monthly (period start plus month-ends). Parameters estimated at t are used only after t (no look-ahead).
- If history is shorter than the requested lookback, the estimator uses all available observations up to the update date (equity starts 2003-12-01; listed-option panels start 2008-01).

V3 empirical study · Instruction (continued)

Six models · three underlyings · four volatility regimes · 1.5-year monthly calibration

Shared market sample and filters (not model-specific)

- For each company $\times$ regime, one listed-call sample is drawn once and reused by all six models: nearest-ATM call on each Monday (next session if Monday is closed), DTE 7–60, listed expiry, as-of that date.
- Estimation filters, applied in order to every quote used for Heston/Heston–Merton calibration and to the LSM comparison set: (1) calls only, finite S , K , C ; (2) no-arbitrage $C \geq \max(0, S-K)$; (3) $7 \leq \text{DTE} \leq 60$; (4) $|S/K - 1| \leq 10\%$; (5) premium ≥ 0.05 , valid bid-ask with relative spread $\leq 50\%$ when those columns exist, volume ≥ 1 when present.
- Return-based models (GBM, Merton, GARCH, GARCH–Merton) never see option quotes in §4; they still price the same filtered LSM contracts in evaluation.

Model-correctness adjustments (V3)

- Heston / Heston–Merton: $(\kappa, \theta, \xi, \rho, v_0)$ are option-implied Fourier NLS (Albrecher little-trap), not Method A realized-variance moments. μ is the P-measure lookback mean of stock returns; LSM replaces $\mu \rightarrow r$.
- Euler ordering: the stock increment over $[t, t+\Delta t]$ uses the current variance v_t ; $v_{t+\Delta t}$ is updated afterwards. No look-ahead in the variance.
- LSM: Longstaff–Schwartz American call on the full path cloud ($n_{\text{paths}} = 10000$, seed 42, $n_{\text{steps}} = \text{DTE}$, $\Delta t = 1/252$). Paths are not averaged before stopping.

Metrics

- Primary: $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}}) / C_{\text{mkt}})^2)}$. Ranking uses this number only.
- Secondary: $\text{MAE} = \text{mean } |C_{\text{model}} - C_{\text{mkt}}|$ (dollars). $\text{Bias \$} = \text{mean}(C_{\text{model}} - C_{\text{mkt}})$ (dollars). $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$. Positive bias = model expensive vs market. Early-exercise fraction = mean share of LSM paths that stop before expiry.

Page 3 · Filters, sample, and no-look-ahead

Shared evaluation sample · n listed calls (identical for all six models)

Regime	Window	SPY n	AAPL n	MSFT n
2008-2009 Crisis	2008-08-01 → 2009-07-31	50	50	50
2013-2014 Normal	2014-01-01 → 2014-12-31	50	50	50
2018-2019 Late-cycle	2018-10-01 → 2019-09-30	50	51	50
2019-2020 COVID	2019-09-01 → 2020-08-31	50	50	50

Filter funnel · last-step n in the 1.5-year lookback ending at each regime (processed call panel)

Ticker	Regime	Input*	No-arb	DTE 7-60	$ S/K-1 \leq 10\%$	Liquidity
SPY	2008-2009	3359	3311	3311	3203	3203
SPY	2013-2014	12815	12548	12548	12405	12405
SPY	2018-2019	35222	35104	35104	34809	34809
SPY	2019-2020	42100	41961	41961	41379	41379
AAPL	2008-2009	685	685	685	652	652
AAPL	2013-2014	8925	8875	8875	8633	8633
AAPL	2018-2019	6309	6297	6297	6084	6084
AAPL	2019-2020	8010	7992	7992	7713	7713
MSFT	2008-2009	566	566	566	538	538
MSFT	2013-2014	3299	3237	3237	3175	3175
MSFT	2018-2019	7814	7752	7752	7583	7583
MSFT	2019-2020	6892	6852	6852	6655	6655

*Input is the processed call panel restricted to (regime end – 1.5-year, regime end]. Processed files already drop many raw quotes; the funnel above is the additional V3 research filter.

No look-ahead. Monthly parameters at update date t use only equity returns and listed quotes with trading_date $\leq t$. LSM prices the Monday contract using the latest calibration row with date $\leq$ that Monday. Quotes are as-of the grid timestamp (same session, else the prior session).

SPY 2008–2009 uses bid-ask mid when the source mark is \$0.01 (a GitHub field error, not missing trades). After that repair the processed panel has weekly coverage, so n matches AAPL/MSFT (~50). All six models still share that exact sample.

Heston / Heston-Merton calibration further subsamples the filtered lookback surface to ≤ 24 contracts (maturity $\times$ moneyness strata). GBM / Merton / GARCH / GARCH-Merton estimate from lookback log returns only (jumps: 3σ residual threshold where applicable).

Page 4 · Table block · SPY · six models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, monthly rolling, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	55.01	1.4054	+0.2705	+21.16	24.5%	6
GARCH	52.86	1.6883	+1.6456	+43.57	20.8%	3
Heston	32.75	1.0960	+0.5803	+20.80	26.9%	2
Merton	52.87	1.3685	+0.1826	+18.56	20.3%	4
GARCH-Merton	54.76	1.7780	+1.7384	+46.28	20.5%	5
Heston-Merton	31.60	1.0435	+0.5375	+19.11	21.4%	1

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	38.08	0.6850	+0.4678	+24.88	26.1%	2
GARCH	38.24	0.6860	+0.4700	+24.96	23.3%	3
Heston	52.46	1.0201	+0.9451	+42.44	26.9%	5
Merton	37.52	0.6728	+0.4520	+24.20	24.7%	1
GARCH-Merton	42.83	0.7744	+0.6153	+30.82	22.5%	4
Heston-Merton	67.16	1.2748	+1.2103	+53.41	24.5%	6

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	49.51	1.6628	+1.1528	+32.90	21.8%	2
GARCH	71.72	2.1416	+1.7421	+45.61	21.9%	3
Heston	73.93	2.3623	+2.1183	+54.85	22.3%	4
Merton	47.62	1.5845	+1.0524	+30.44	16.2%	1
GARCH-Merton	77.02	2.4122	+2.1360	+54.80	20.5%	5
Heston-Merton	84.01	2.7641	+2.5349	+64.41	17.0%	6

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	64.88	3.6003	+1.6141	+44.13	23.9%	3
GARCH	56.59	2.9250	+2.0268	+34.97	32.7%	1
Heston	65.44	3.7704	+2.6179	+50.30	26.4%	4
Merton	60.38	3.3441	+1.2734	+38.99	18.7%	2
GARCH-Merton	66.14	3.4255	+2.8053	+46.06	31.4%	5
Heston-Merton	66.64	3.5171	+2.3874	+47.68	15.7%	6

Page 5 · Table block · AAPL · six models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, monthly rolling, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	57.67	2.4326	+1.3849	+31.10	22.8%	5
GARCH	48.71	2.7044	+2.5500	+40.72	19.1%	3
Heston	31.71	1.5996	+1.0721	+21.09	22.7%	1
Merton	55.67	2.3640	+1.2255	+28.42	18.3%	4
GARCH-Merton	58.94	3.2224	+3.1062	+50.39	15.9%	6
Heston-Merton	32.30	1.6138	+1.0704	+21.37	18.5%	2

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	50.20	3.5766	+3.5387	+38.63	24.6%	5
GARCH	45.44	3.1849	+3.0454	+29.67	22.2%	3
Heston	37.85	2.5745	+2.5702	+29.62	20.8%	2
Merton	45.50	3.2302	+3.1799	+33.94	19.5%	4
GARCH-Merton	65.87	4.9053	+4.9027	+53.60	17.8%	6
Heston-Merton	31.91	1.9670	+1.9420	+21.53	15.2%	1

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	32.53	1.7033	+1.0081	+18.76	23.9%	2
GARCH	48.45	2.0164	+1.5757	+29.40	20.0%	5
Heston	45.14	2.3820	+2.2976	+38.59	21.0%	4
Merton	30.59	1.5914	+0.8543	+16.31	19.9%	1
GARCH-Merton	59.31	2.6521	+2.3998	+42.76	18.5%	6
Heston-Merton	41.69	2.0875	+1.9310	+32.87	18.2%	3

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	46.19	3.9995	+2.3663	+30.76	23.8%	4
GARCH	61.17	4.8546	+4.4651	+43.67	24.2%	5
Heston	42.24	3.7296	+2.9444	+33.15	23.3%	2
Merton	43.61	3.6886	+1.9753	+27.36	18.9%	3
GARCH-Merton	71.81	5.9515	+5.7532	+57.54	24.2%	6
Heston-Merton	40.88	3.4396	+2.5150	+30.03	16.7%	1

Page 6 · Table block · MSFT · six models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, monthly rolling, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	44.02	0.3723	+0.0897	+17.99	24.1%	4
GARCH	70.79	0.5879	+0.5440	+54.27	30.0%	5
Heston	26.51	0.2560	+0.0897	+12.61	24.7%	1
Merton	42.94	0.3684	+0.0716	+16.18	19.7%	3
GARCH-Merton	81.24	0.6935	+0.6704	+66.90	29.6%	6
Heston-Merton	26.84	0.2552	+0.0910	+12.75	18.6%	2

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	62.49	0.4283	+0.4261	+52.98	24.7%	4
GARCH	90.24	0.6511	+0.6511	+78.36	25.2%	5
Heston	31.79	0.1889	+0.1565	+21.39	22.6%	2
Merton	54.85	0.3679	+0.3602	+45.39	17.1%	3
GARCH-Merton	112.93	0.8656	+0.8656	+102.63	21.6%	6
Heston-Merton	26.54	0.1648	+0.1181	+16.70	12.2%	1

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	41.73	1.0839	+0.8054	+27.55	23.0%	4
GARCH	42.43	1.0703	+0.9006	+30.13	18.9%	5
Heston	37.47	0.9577	+0.8192	+26.22	21.8%	2
Merton	40.33	1.0504	+0.7474	+25.77	19.8%	3
GARCH-Merton	52.89	1.3664	+1.2944	+42.08	17.0%	6
Heston-Merton	33.23	0.7984	+0.6200	+19.90	16.2%	1

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	45.93	2.0118	+1.0018	+28.12	24.5%	4
GARCH	60.29	2.3979	+1.9268	+35.85	29.6%	5
Heston	40.67	2.0782	+1.5079	+30.43	23.8%	2
Merton	44.13	1.8737	+0.7910	+25.02	19.7%	3
GARCH-Merton	346.80	6.7038	+6.3160	+88.26	28.7%	6
Heston-Merton	35.19	1.7711	+1.0681	+20.23	16.0%	1

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 12 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 12 cells best
1	Heston	43.16	39.26	2
2	Heston-Merton	43.16	34.21	6
3	Merton	46.33	44.82	3
4	GBM	49.02	47.85	0
5	GARCH	57.25	54.73	1
6	GARCH-Merton	90.88	66.00	0

Table S2 · Wins by company (lowest RMSE% in that company×regime)

Model	SPY wins	AAPL wins	MSFT wins	Total
GBM	0	0	0	0
GARCH	1	0	0	1
Heston	0	1	1	2
Merton	2	1	0	3
GARCH-Merton	0	0	0	0
Heston-Merton	1	2	3	6

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
GBM	52.23	50.26	41.25	52.33	49.02
GARCH	57.45	57.97	54.20	59.35	57.25
Heston	30.32	40.70	52.18	49.45	43.16
Merton	50.49	45.96	39.51	49.38	46.33
GARCH-Merton	64.98	73.88	63.07	161.58	90.88
Heston-Merton	30.24	41.87	52.98	47.57	43.16

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
GBM	51.87	52.26	0
GARCH	54.85	54.73	1
Heston	56.15	58.95	0
Merton	49.60	50.24	2
GARCH-Merton	60.19	60.45	0
Heston-Merton	62.35	66.90	1

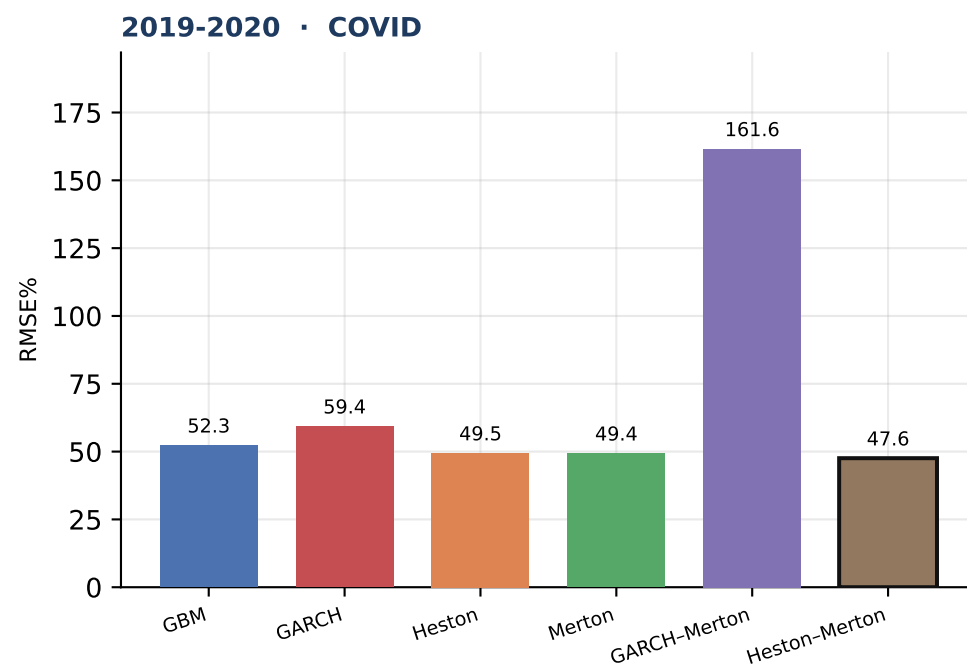
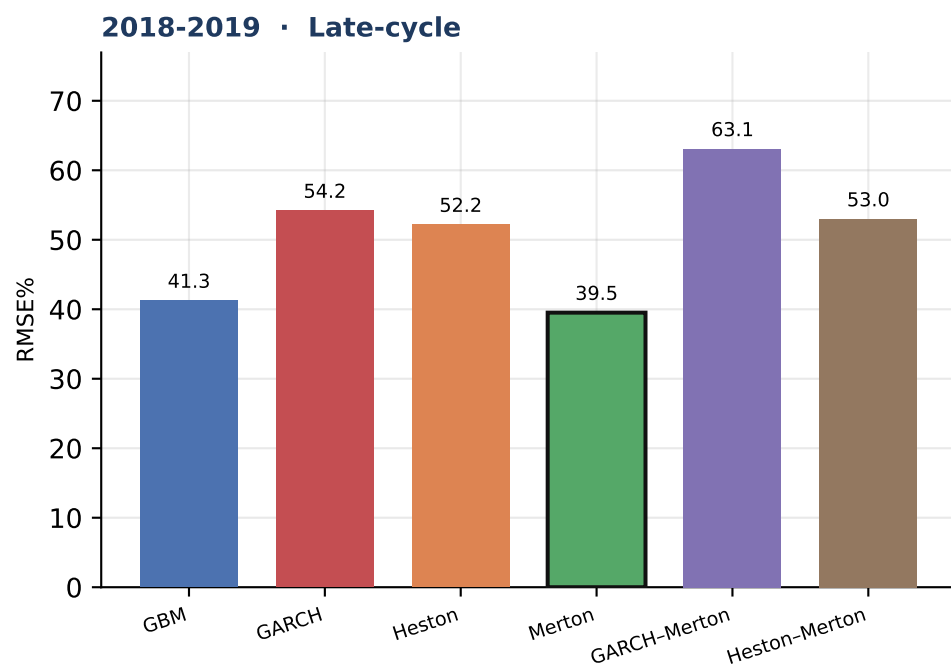
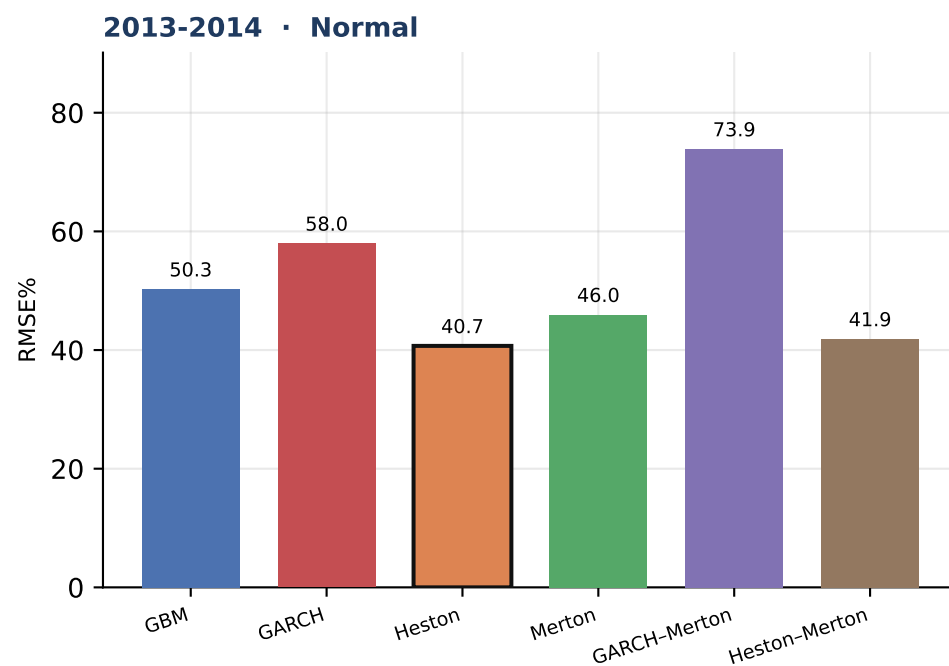
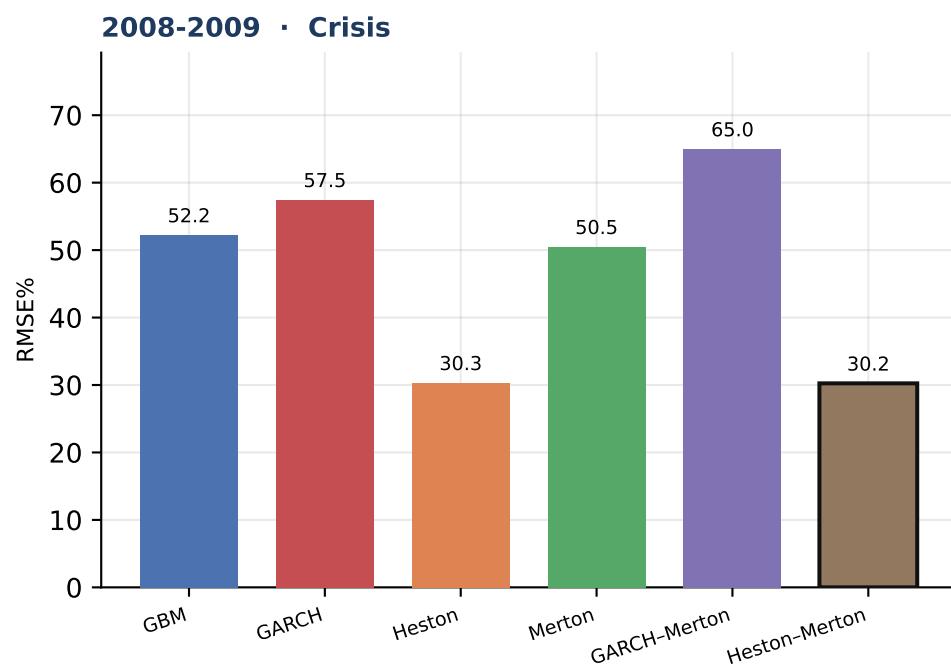
Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
GBM	46.65	48.20	0
GARCH	50.94	48.58	0
Heston	39.23	40.05	1
Merton	43.84	44.56	1
GARCH-Merton	63.98	62.59	0
Heston-Merton	36.69	36.59	2

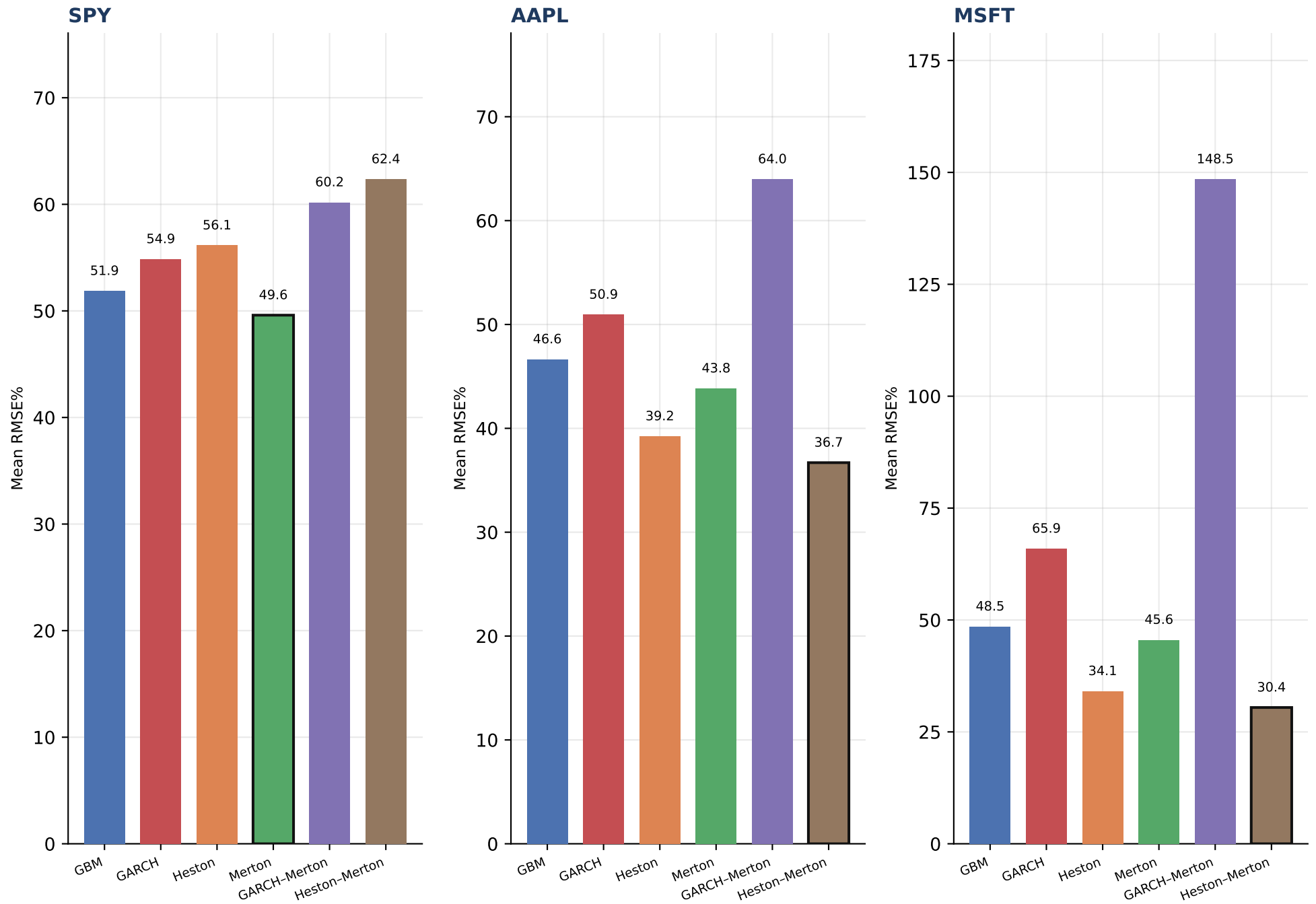
Table S4.3 · MSFT

Model	Mean RMSE%	Median	# regimes best
GBM	48.54	44.98	0
GARCH	65.94	65.54	0
Heston	34.11	34.63	1
Merton	45.56	43.54	0
GARCH-Merton	148.47	97.09	0
Heston-Merton	30.45	30.03	3

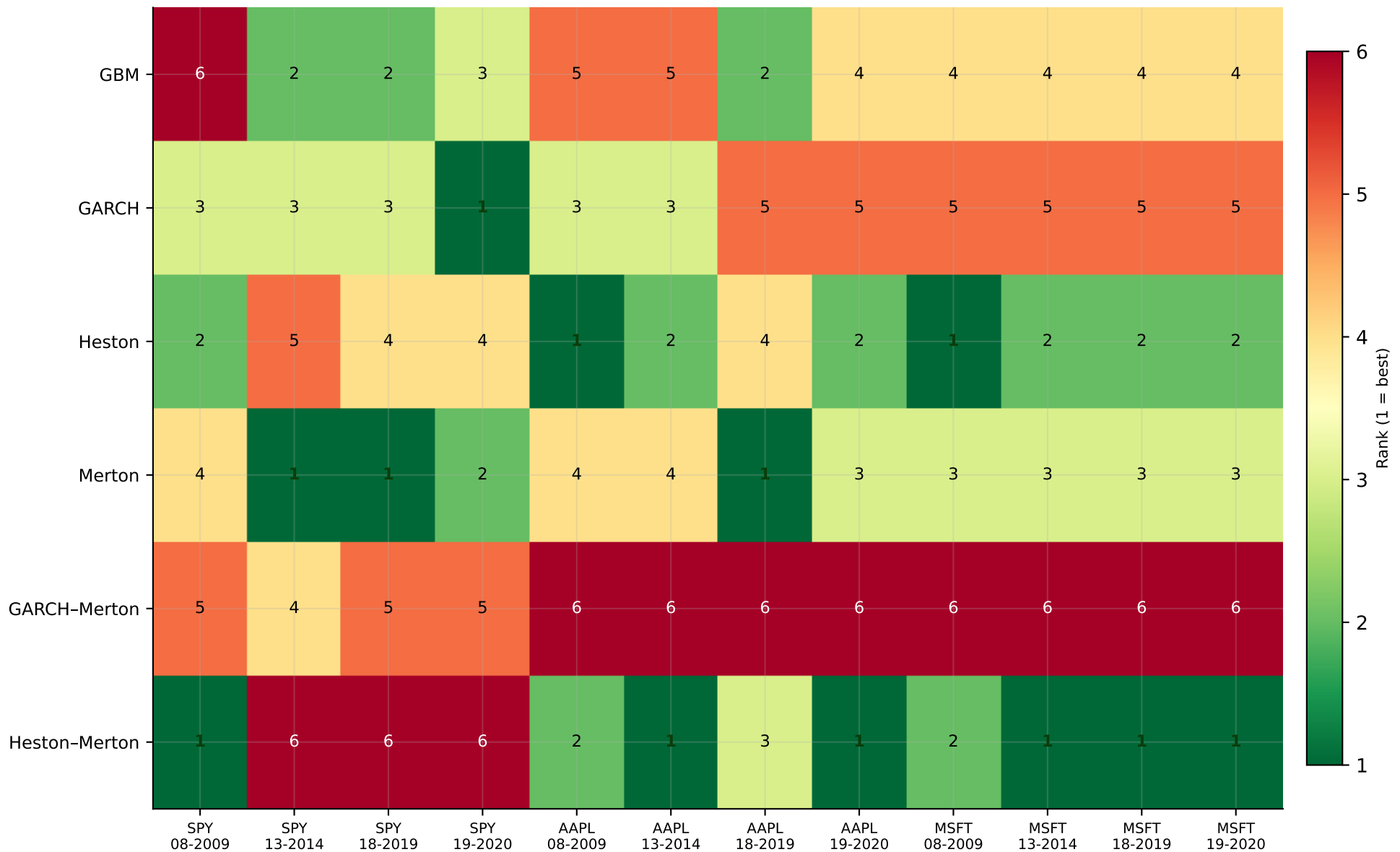
Each column is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.



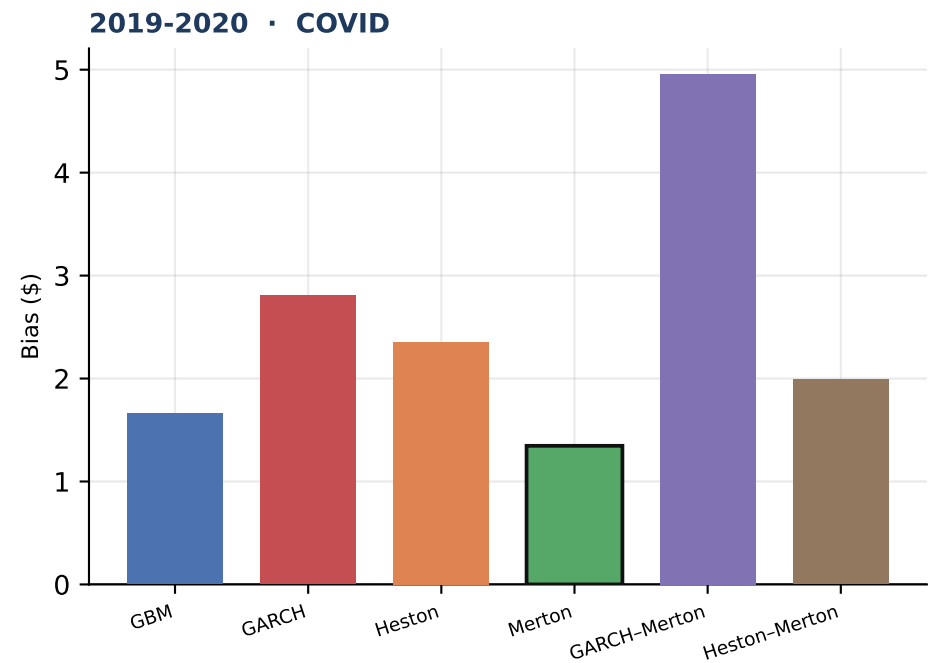
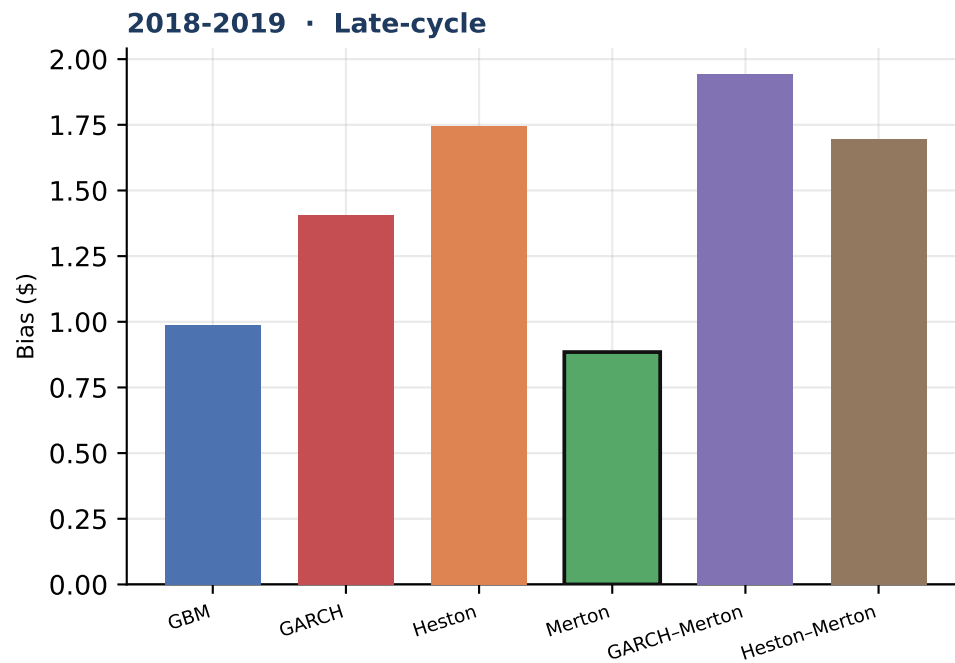
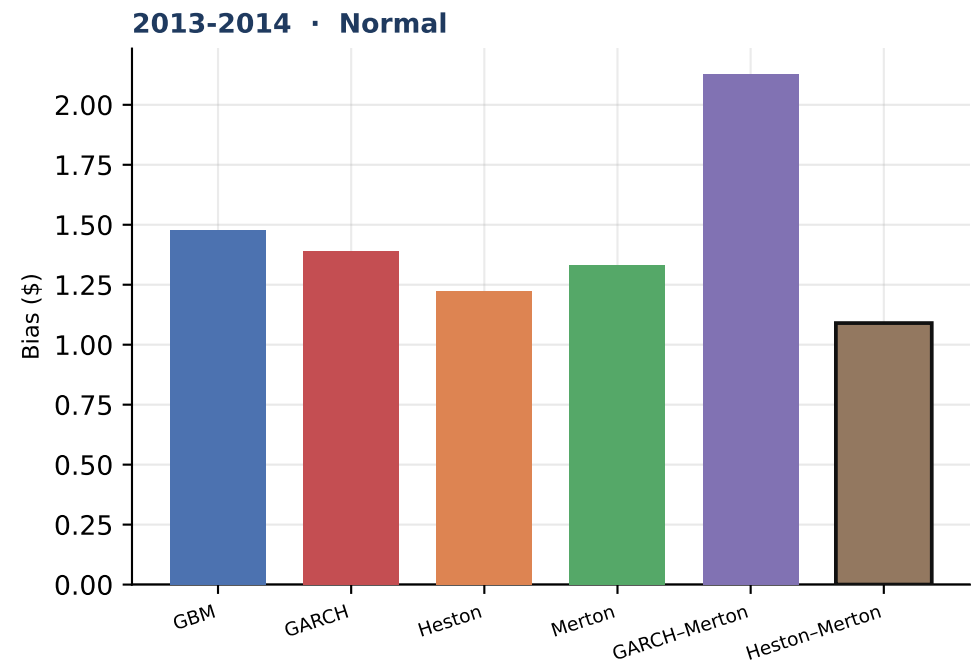
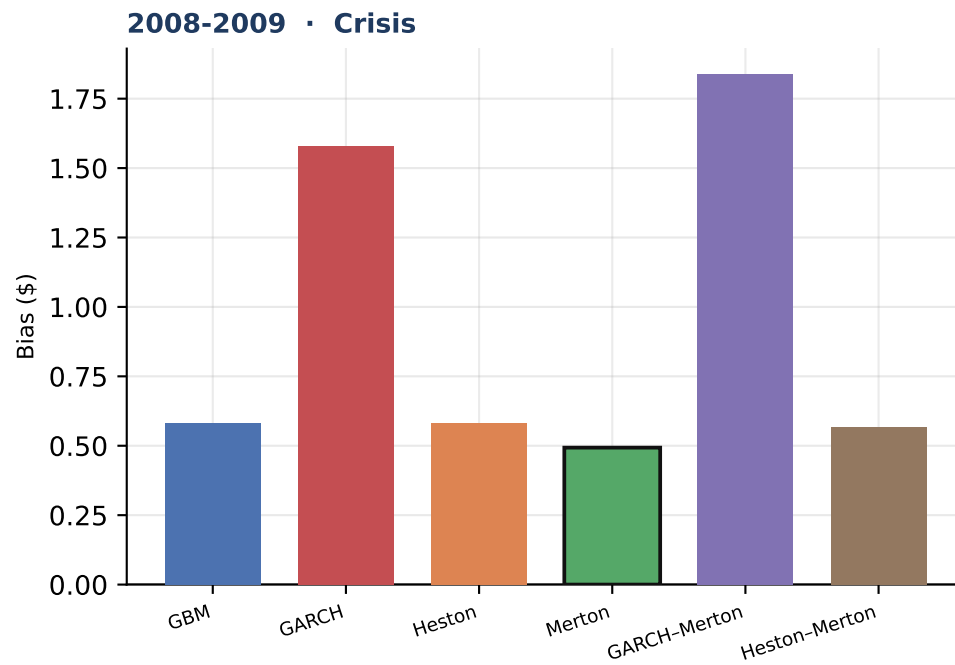
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the three companies.



Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero bias (not the RMSE ranking). Positive = model expensive vs listed quote.

Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 12 company×regime cells, Heston is the best overall model (mean RMSE% = 43.16; 2 of 12 cells).
- Second overall is Heston-Merton (mean RMSE% = 43.16; 6 cells).
- The ranking is not stable. Best-in-cell models across the 12 tables: GARCH, Heston, Heston-Merton, Merton.
- SPY: lowest mean RMSE% is Merton (49.60); wins 2 of 4 regimes.
- AAPL: lowest mean RMSE% is Heston-Merton (36.69); wins 2 of 4 regimes.
- MSFT: lowest mean RMSE% is Heston-Merton (30.45); wins 3 of 4 regimes.
- 2008-2009 (Crisis): best mean RMSE% is Heston-Merton (30.24).
- 2013-2014 (Normal): best mean RMSE% is Heston (40.70).
- 2018-2019 (Late-cycle): best mean RMSE% is Merton (39.51).
- 2019-2020 (COVID): best mean RMSE% is Heston-Merton (47.57).
- No-jump models as a group have lower mean RMSE% (49.81) than the three jump models (60.13).
- For the overall winner (Heston), crisis RMSE% is 30.32 vs 40.70 in the 2014 normal year — absolute pricing error is regime-dependent even when the ranking is not.
- These conclusions are conditional on the V3 filters, 1.5-year monthly calibration, Monday ATM sample, LSM with 10,000 paths, and American-call quotes only. They are not a statement about European options or other tenors.

How to read the 12 detailed tables

- Each of Tables SPY-1...MSFT-4 uses exactly the same option contracts for all six models. Differences are the dynamics, not the sample.
- BEST is always lowest RMSE%. MAE, Bias \$, Bias%, and early-exercise are reported, not used for the ranking.
- A model that is BEST on RMSE% can still be biased (systematically expensive or cheap).
- Early-exercise fractions are a model implication, not an accuracy score; American calls on these tenors often have modest early-exercise value.